

ENBC455 - HW11.py > [🔍] f

```
1  # Name: Christopher A. Lee
2  # Date: 04/28/2026
3  # Prof: Dr. Callura
4  # Course: Bioinformatics - ENBC 4553
5
6  # Libraries
7  from Bio import SeqIO
8
9  # Problem 1
10 with open('hw11_1.txt', 'r') as f:
11     contigs = [str(i.strip()) for i in f.readlines()]
12     contigs_lengths = [len(contig) for contig in contigs]
13     contigs_sorted = sorted(contigs_lengths, reverse=True)
14
15     N = 0
16     N50 = 0
17     N75 = 0
18     total = sum(contigs_sorted)
19
20     for i in contigs_sorted:
21         N += i
22         if N50 == 0 and N >= (total * 0.5):
23             N50 = i
24         if N75 == 0 and N >= (total * 0.75):
25             N75 = i
26         break
27
28     print(N50, N75)
29
```

175

137

ENBC455-HW11-Problem3.R

Source on Save

Run

Source

```
1 # Packages
2 install.packages("BiocManager")
3 BiocManager::install(c("GenomicFeatures", "AnnotationDbi", "ballgown"))
4
5 # Ensure the library and data directory are loaded
6 library(ballgown)
7 data_directory = system.file('extdata', package='ballgown')
8
9 # Create the ballgown object (Loads your data into R)
10 bg = ballgown(dataDir=data_directory, samplePattern='sample', meas='all')
11
12 # Define the experimental groups (0 = Wild-Type, 1 = Engineered)
13 pData(bg) = data.frame(id=sampleNames(bg), group=c(0,0,1,1))
14
15 # Run the statistical test for differential expression
16 stat_results = stattest(bg, feature='transcript', meas='FPKM', covariate='group')
17
18 # Filter the results for statistically significant DEGs (p-value < 0.05)
19 sig_results = subset(stat_results, stat_results$pval < 0.05)
20
21 # Number of DEGs
22 nrow(sig_results)
23
24 # Export the results to a TSV file
25 write.table(sig_results, "outputfilename.tsv", sep="\t", quote=FALSE, row.names=FALSE)
```

19:61 (Top Level) R Script

Environment

History

Connections

Tutorial

Import Dataset

1.12 GiB

List

R

Global Environment

Data

bg	Large ballgown (1 MB)
sig_results	6 obs. of 4 variables
\$ feature: chr "transcript" "transcript" "transcript" "transcript" ...	
\$ id : chr "10" "123" "231" "695" ...	
\$ pval : num 0.00762 0.02035 0.00586 0.03296 0.00629 ...	
\$ qval : num 0.251 0.445 0.251 0.544 0.251 ...	
stat_results	100 obs. of 4 variables
\$ feature: chr "transcript" "transcript" "transcript" "transcript" ...	
\$ id : chr "10" "25" "35" "41" ...	
\$ pval : num 0.00762 0.64206 0.36671 0.72534 0.16447 ...	
\$ qval : num 0.251 0.836 0.732 0.898 0.732 ...	

Values

data_directory	"/Library/Frameworks/R.framework/Versions/4.5-arm64/Re...
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Console

Terminal

Background Jobs

R 4.5.1 · ~/

```
> # Number of DEGs
> nrow(sig_results)
[1] 6
>
> # Export the results to a TSV file
> write.table(sig_results, "outputfilename.tsv", sep="\t", quote=FALSE, row.names=FALSE)
> install.packages("BiocManager")
trying URL 'https://cran.rstudio.com/bin/macosx/big-sur-arm64/contrib/4.5/BiocManager_1.30.27.tgz'
Content type 'application/x-gzip' length 660755 bytes (645 KB)
=====
downloaded 645 KB

The downloaded binary packages are in
  /var/folders/tn/tj4_th0577vb43zq_58mwbfm0000gn/T//RtmpTR4cZY/downloaded_packages
> |
```

Files

Plots

Packages

Help

Viewer

Presentation

Zoom

Export